

55) Find a compound proposition logically equivalent to $(p \rightarrow \neg q)$ using only the logical operators $\downarrow$.

Soln: $p \rightarrow \neg q \equiv \neg p \vee q$ (Conditional disjunction equivalence)
 $\equiv (\neg p \downarrow q) \downarrow (\neg p \downarrow q)$ (Ex-54(b))
 $\equiv (p \downarrow p) \downarrow (p \downarrow p) \downarrow q$ (Ex-54(a))

56) Show that $\{\downarrow\}$ is a functionally complete collection of logical operators.

Soln: Let p and q be compound propositions.

$$\begin{aligned} p \wedge q &\equiv \neg(p \downarrow q) \text{ (By Ex-51)} \\ &\equiv \neg p \vee \neg q \text{ (By 1st De Morgan's Law)} \\ &\equiv (\neg p \downarrow \neg q) \downarrow (\neg p \downarrow \neg q) \text{ (By Ex-54(b))} \\ &\equiv ((p \downarrow p) \downarrow (q \downarrow q)) \downarrow ((p \downarrow p) \downarrow (q \downarrow q)) \text{ (By Ex-54(a))} \end{aligned}$$

I had to prove the reverse.

By Ex-54(c) $\{\downarrow\}$ is functionally complete.

$$\begin{aligned} p \downarrow q &\equiv \neg(p \vee q) \text{ (By Ex-53)} \\ &\equiv \neg p \wedge \neg q \text{ (By 2nd De Morgan's Law)} \\ &\equiv \neg(\neg(\neg p \wedge \neg q)) \text{ (Double-negation Law)} \\ &\equiv \neg(\neg p \downarrow \neg q) \text{ (By Ex-51)} \\ &\equiv \neg(\neg p \downarrow \neg q) \end{aligned}$$

When $p \downarrow q$ is True, by defn. of NAND, $p \downarrow q$ is False. By defn. of negation, $\neg p \downarrow \neg q$ is True. $\therefore p \downarrow q$ and $\neg p \downarrow \neg q$ are equivalent.

$$\begin{aligned} \therefore p \downarrow q &\equiv \neg(\neg p \downarrow \neg q) \equiv \neg((p \downarrow p) \downarrow (q \downarrow q)) \text{ (Since } p \downarrow p \text{ and } \neg p \text{ are equivalent)} \\ &\equiv ((p \downarrow p) \downarrow (q \downarrow q)) \downarrow ((p \downarrow p) \downarrow (q \downarrow q)) \end{aligned}$$

$\therefore$ Any expression having $\downarrow$ can be rewritten using only $\downarrow$

$\therefore \{\downarrow\}$ is a functionally complete collection of logical operators $\square$

57) Show that $p \downarrow q$ and $q \downarrow p$ are equivalent.

Soln: $p \downarrow q \equiv \neg(p \wedge q)$ (By Ex-51)
 $\equiv \neg(q \wedge p)$ (By 2nd Commutative Law)
 $\equiv q \downarrow p$ (By Ex-51)